

RATIONAL ACTIVATIONS AND LOSS-INFORMED OPTIMIZATION FOR TRANSFORMERS

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Paper under double-blind review

ABSTRACT

Learning an activation function changes both the represented features and their response to parameter updates. We introduce full TILLER (Tangents Informed by a Loss Ledger for Equal-budget Reweighting) and the hybrid TILLER–Muon, two loss-informed training methods for Transformers. GRAIN, Groupwise Rational Activation with Internal Normalization, learns rational responses in normalized feature coordinates and restores each group’s input scale at its output. The shared TILLER update uses these learned responses to construct and coordinate matrix directions. It rotates Muon directions toward factorized adaptive tangents, then allocates their combined norm across groups using a compact history of directional loss responses. This allocation accounts for interactions among updates across the network. Full TILLER maintains this coordination throughout training; TILLER–Muon combines it with momentum-preserving Muon continuation. Both use learned nonlinearities to represent features and guide structured updates according to their joint effect on the loss.

1 INTRODUCTION

The feed-forward blocks of a Transformer determine how intermediate features are transformed and recombined. Their nonlinear responses are commonly fixed in advance, as in GELU and the SiLU gate of SwiGLU (Hendrycks & Gimpel, 2016; Shazeer, 2020). Learning these responses gives the network an additional way to adapt its representation. Trainable rational functions provide a compact family for this purpose: they can approximate familiar activations and learn new shapes through the same gradient updates used to train the surrounding weights (Molina et al., 2020). Their compositional structure also supports approximation advantages over fixed activation families (Boullé et al., 2020; Tang & Townsend, 2026). Group-shared rational functions have already made this approach practical in Transformer feed-forward layers (Yang & Wang, 2025).

A learned nonlinearity also changes the optimization problem. An update to the incoming projection reaches the block output through the current activation Jacobian; an update to the outgoing projection acts on the current activated features. As the rational coefficients change, both responses change with them. Consequently, matrix updates of the same magnitude can produce different feature changes at different groups or stages of training. Updates from separate groups can also reinforce or offset one another in their effect on the loss. These relationships make the activation response useful information for choosing an update, beyond its role in the forward computation.

Matrix-aware optimizers offer a natural starting point for using this information. Shampoo preconditioning gradients according to their tensor structure (Gupta et al., 2018), while Muon approximately orthogonalizes momentum updates and has been developed for language-model training (Jordan et al., 2024; Liu et al., 2025). Norm-based interpretations clarify the geometry of such updates (Bernstein & Newhouse, 2024). Activation and derivative statistics also enter curvature-based methods such as K-FAC and the matrix surrogate of Newton–Muon (Martens & Grosse, 2015; Du & Su, 2026). Our focus is the connection between learned nonlinear responses and the joint effect of structured update directions. We use that connection at two levels: to adjust the direction within a matrix block and to coordinate the strengths of those directions across the network.

We develop full TILLER (Tangents Informed by a Loss Ledger for Equal-budget Reweighting) and the hybrid TILLER–Muon around a shared activation and loss-informed update construction. The activation, GRAIN (Groupwise Rational Activation with Internal Normalization), evaluates a separate

learned rational function for each feature group in coordinates normalized by the group’s root mean square. It then restores that scale, allowing the rational coefficients to control response shape while the output retains the input amplitude. This construction supplies a structured Jacobian that describes the feature changes induced by incoming updates. The activated features provide the corresponding response for outgoing updates.

The shared TILLER update uses these responses to guide optimization. Within each matrix group, it constructs a norm-preserving rotation from a Muon direction toward a factorized adaptive direction. Response participation and departure from the initial response determine the rotation. Across groups, sampled directional loss responses form a compact temporal ledger. A quadratic model combines the current descent signal with the interactions recorded in this ledger to redistribute a fixed total update norm. The optimizer can therefore change where the structured update acts while retaining its overall magnitude.

These components provide a joint design for learning nonlinear representations and optimizing the matrices around them. Full TILLER carries response-guided coordination through training. TILLER–Muon combines this coordination with ordinary Muon continuation, preserving the learned activation, matrix momentum, and the optimizer state of the remaining parameters. The two variants share the same activation and update construction; their schedules determine how long loss-response information guides the matrix directions.

2 RELATED WORK

Learnable activations extend the function family available to a network beyond a fixed scalar response. PReLU learns a negative-input slope (He et al., 2015), whereas GELU and SwiGLU specify smooth responses that have become common choices for feed-forward computation (Hendrycks & Gimpel, 2016; Shazeer, 2020). Padé activation units learn rational coefficients end to end and use a modified denominator to avoid real poles (Molina et al., 2020). Rational neural networks connect the composition of low-degree activations to approximation efficiency (Boullé et al., 2020). The paper *Rational Neural Networks have Expressivity Advantages* extends this approximation-theoretic perspective to modern smooth activation families and gated and Transformer-style nonlinearities (Tang & Townsend, 2026). Group-shared rational functions in the Kolmogorov–Arnold Transformer provide the closest architectural precedent for learning distinct responses within a feed-forward block (Yang & Wang, 2025). GRAIN uses an amplitude-restoring normalization within each learned rational response.

Normalization determines the coordinates in which an activation is learned. Layer normalization centers and rescales within-example features (Ba et al., 2016), while RMS normalization uses their root mean square without mean centering (Zhang & Sennrich, 2019). For trainable rational activations, the interaction also involves the coefficients themselves: affine changes before a rational function can be absorbed into a coefficient reparameterization (Tang & Townsend, 2026). GRAIN ties the normalization scale to the current feature group and restores it after the rational map. Its derivative therefore combines the elementwise rational derivative with the response induced by a change in group scale. TILLER uses both contributions when relating incoming matrix directions to feature changes.

Adaptive optimizers use gradient statistics to transform update directions. AdamW separates parameter shrinkage from the adaptive gradient step (Loshchilov & Hutter, 2019); Adafactor approximates gradient second moments with row and column factors (Shazeer & Stern, 2018), and CAME adds confidence-guided scaling to factorized adaptation (Luo et al., 2023). Shampoo uses tensor-structured preconditioners (Gupta et al., 2018), and SOAP applies Adam in the corresponding preconditioner eigenbasis (Vyas et al., 2025). Muon instead transforms matrix momentum through approximate orthogonalization (Jordan et al., 2024), with connections to spectral-norm geometry (Bernstein & Newhouse, 2024) and extensions to larger language-model training (Liu et al., 2025). TILLER constructs its local directions from Muon and factorized adaptive updates, with the learned activation response controlling a rotation between them.

Activation-informed optimization and low-dimensional search provide further context for this construction. K-FAC approximates layerwise Fisher blocks from activations and backpropagated derivatives (Martens & Grosse, 2015). Newton–Muon uses a matrix quadratic surrogate to introduce input-second-moment preconditioning (Du & Su, 2026). Krylov Subspace Descent optimizes over

a subspace generated from gradient and curvature information (Vinyals & Povey, 2012). TILLER coordinates coefficients of its selected group directions through their sampled loss responses. Its temporal response metric uses deterministic spectral sketching from Frequent Directions and the isotropic correction of Robust Frequent Directions (Ghashami et al., 2016; Luo et al., 2019). The resulting quadratic model couples directions across groups through their joint loss effects.

Training schedules offer another way to combine optimization behavior. SWATS transitions from Adam to SGD using a projection-based trigger (Keskar & Socher, 2017); Schedule-Free optimization uses iterate averaging to alter the role of learning-rate scheduling (Defazio et al., 2024). TILLER–Muon combines response-guided coordination with matrix-level Muon continuation, transferring momentum while retaining GRAIN and the AdamW state.

3 METHOD

Full TILLER and TILLER–Muon share a response-guided update construction for Transformers with GRAIN activations. Full TILLER uses this construction throughout training; TILLER–Muon combines it with a state-preserving Muon continuation. We first define their common activation, then develop the update rule and the two training schedules.

3.1 GRAIN: GROUPWISE RATIONAL ACTIVATIONS

GRAIN (Groupwise Rational Activation with Internal Normalization) gives each feature group a learned nonlinear response. It separates response shape from input amplitude: a rational function acts on normalized features, and the group scale is restored at its output. For an input $x \in \mathbb{R}^d$, split the intermediate features into G groups of width w , with incoming and outgoing projections $A_g \in \mathbb{R}^{w \times d}$ and $B_g \in \mathbb{R}^{d \times w}$. The block output is

$$y = \sum_{g=1}^G B_g h_g, \quad h_g = \rho_g \phi_g(u_g), \quad u_g = \frac{A_g x}{\rho_g}, \quad \rho_g = \sqrt{\|A_g x\|_2^2 / w + \epsilon_{\text{act}}}. \quad (1)$$

Here ρ_g is the stabilized group root mean square (RMS), u_g contains the normalized features, and ϕ_g acts elementwise:

$$\phi_g(u) = \frac{\sum_{j=0}^5 a_{gj} u^j}{1 + \sum_{j=1}^4 |b_{gj}| |u|^j}. \quad (2)$$

The learned coefficients a_{gj}, b_{gj} control the response shape (Molina et al., 2020; Boullé et al., 2020), and the positive denominator excludes real poles. Internal RMS normalization (Zhang & Sennrich, 2019) lets the rational function operate on relative feature magnitudes as their absolute scale changes. Restoring ρ_g preserves amplitude information for the outgoing projection. All groups start from a common SiLU approximation and subsequently learn separate shapes. With $Gw = 3H/2$, the two projections contain the same number of dense weights as a SwiGLU block of intermediate width H (Shazeer, 2020).

The two projections expose complementary information about a proposed update. An incoming direction changes the features through the activation Jacobian $J_g = \partial h_g / \partial (A_g x)$; an outgoing direction acts directly on h_g . TILLER uses both paths to assess a direction in the context of the learned nonlinearity. The Jacobian and its interpretation are derived in Appendix A.

3.2 FULL AND HYBRID TILLER OPTIMIZATION

TILLER (Tangents Informed by a Loss Ledger for Equal-budget Reweighting) makes two linked choices: the orientation of each matrix update and the strengths with which group updates act together. Let $i = (\ell, g)$ index the $C = LG$ layer–group pairs in a network of depth L . For either projection role, form a Muon direction p_i and a factorized adaptive direction a_i from the same momentum (Jordan et al., 2024; Liu et al., 2025; Shazeer & Stern, 2018). The former rebalances matrix singular values; the latter uses row and column gradient histories to rescale coordinates. Removing the component of a_i parallel to p_i gives $a_i^\perp = a_i - \langle a_i, p_i \rangle p_i / \|p_i\|_F^2$. This orthogonal component provides an alternative orientation without changing the norm of the parent direction.

The activation response sets the amount of rotation. A participation statistic π_i measures the breadth of the response, using the singular values of J_i for an incoming update and the coordinates of h_i for an outgoing update. A congruence statistic κ_i compares the learned response with its initialization at the same current inputs. Both emphasize positions with greater loss sensitivity. The fraction $e_i = \pi_i(1 - \kappa_i^2)$ allocates more direction energy to the tangent when a broad response has departed from its initial geometry. The candidate is

$$\tilde{q}_i = \sqrt{1 - e_i} p_i + \sigma_i \sqrt{e_i} \frac{\|p_i\|_F}{\|a_i^\perp\|_F} a_i^\perp, \quad \sigma_i = \text{sign} \langle g_i, a_i^\perp \rangle. \quad (3)$$

Here g_i is the clipped gradient, $\text{sign}(0) = 1$, and $0 \leq e_i \leq 1$. The sign orients the tangent toward descent; orthogonality gives $\|\tilde{q}_i\|_F = \|p_i\|_F$. The selected direction q_i is the candidate when both it and the parent have positive gradient contraction, and is the parent otherwise. Appendix B specifies the statistics and selection.

The resulting directions q_i^A, q_i^B are then coordinated through their loss responses. At a shared example–token position k , let $x_{k\ell}$ be the block input, h_{ki} its group features, and $\xi_{k\ell}$ the downstream loss derivative at the block output. The scalar response of the paired group direction is

$$S_{ki} = \langle \xi_{k\ell}, B_i J_{ki} q_i^A x_{k\ell} + q_i^B h_{ki} \rangle. \quad (4)$$

Each column of S describes one group across the same probes. For a coefficient vector c , Sc is their combined response. Its squared norm captures reinforcement and cancellation between group contributions, including contributions from different layers. TILLER retains a discounted history of these measurements in a loss ledger. A Robust Frequent Directions sketch preserves dominant response combinations and an isotropic remainder (Ghashami et al., 2016; Luo et al., 2019).

Let G_t be the resulting response metric, r_t its cross term with decoupled weight decay, and $s_{t,i} = \langle g_i^A, q_i^A \rangle + \langle g_i^B, q_i^B \rangle$ the current minibatch descent signal. At learning rate η , the coefficients are chosen using the quadratic surrogate

$$F_t(c) = -\eta s_t^\top c + \frac{\eta^2}{2} (c^\top G_t c + 2r_t^\top c), \quad \sum_i \omega_i c_i^2 = \sum_i \omega_i, \quad \omega_i = \|q_i^A\|_F^2 + \|q_i^B\|_F^2. \quad (5)$$

The linear term rewards current descent; the quadratic term measures the joint response through the ledger. The constraint redistributes a fixed squared direction norm across groups. Coefficients can strengthen, attenuate, or reverse a group according to its contribution to the joint proposal. The metric is applied through its sketch factor, and a projected solver selects coefficients that improve this surrogate over $c = 1$ while preserving layerwise descent.

The selected coefficients update $A_i \leftarrow (1 - \eta \lambda_{\text{wd}}) A_i - \eta c_i q_i^A$ and analogously for B_i , where λ_{wd} is weight decay. Attention uses the same response-guided rotation with layerwise statistics. Rational coefficients and the remaining parameter classes use AdamW (Loshchilov & Hutter, 2019).

Let T be the training horizon and $T_s \in \{1, \dots, T\}$ the duration of loss-informed coordination. Full TILLER uses $T_s = T$. TILLER–Muon uses $T_s < T$: it applies TILLER for the first T_s updates and then continues with ordinary Muon on the full feed-forward and attention matrices. The transition transfers the first-moment buffers, retains the AdamW state, and continues the same learning-rate schedule. GRAIN and its learned coefficients remain active throughout. The hybrid reduces ongoing coordination work while retaining the learned representation and gradient history. It provides a time-saving continuation when later-stage gains from further coordination are small. The switch point can be chosen for the training regime. Table 1 gives both branches in a single algorithm, with the ledger and coefficient solver derived in Appendix C.

4 EXPERIMENTS

[TODO: Experiment section]

5 DISCUSSION

6 CONCLUSION

AI USE STATEMENT

Generative AI tools assisted with code implementation, mathematical cross-checking, literature retrieval, and manuscript drafting and revision.

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A NORMALIZED RATIONAL RESPONSE GEOMETRY

GRAIN learns the shape of a nonlinearity in coordinates that follow the scale of its inputs. The group normalization and scale restoration act together: the first controls the inputs presented to the rational function, while the second carries amplitude information into the outgoing projection. This construction also exposes the response geometry used by TILLER.

For one group, write $z = Ax \in \mathbb{R}^w$ and suppress its layer and group indices. The activation is

$$\Phi(z) = \rho\phi(u), \quad u = z/\rho, \quad \rho = (z^\top z/w + \epsilon_{\text{act}})^{1/2}. \quad (6)$$

The rational coefficients specify a response to relative feature magnitudes. In the idealized case $\epsilon_{\text{act}} = 0$ and $z \neq 0$, positive rescaling leaves u unchanged and gives $\Phi(az) = a\Phi(z)$ for $a > 0$. The positive stabilizer makes the map defined at zero. It is therefore useful to view GRAIN as a learned shape together with a data-dependent amplitude, rather than as a rational function applied to unnormalized features.

An incoming matrix update changes both of these quantities. The normalization derivatives are

$$\nabla_z \rho = u/w, \quad \frac{\partial u}{\partial z} = \rho^{-1} \left(I - \frac{uu^\top}{w} \right). \quad (7)$$

With $f = \phi(u)$, $d_\phi = \phi'(u)$, and $r_\phi = f - u \odot d_\phi$, the product rule yields

$$J = \frac{\partial \Phi}{\partial z} = \frac{fu^\top}{w} + \text{Diag}(d_\phi) \left(I - \frac{uu^\top}{w} \right) = \text{Diag}(d_\phi) + \frac{r_\phi u^\top}{w}. \quad (8)$$

The diagonal term applies the local slope to each coordinate. The rank-one term accounts for their shared change in scale. For a linear response $\phi(u) = au$, $r_\phi = 0$ and $J = aI$: normalization and restoration cancel exactly. For a nonlinear response, r_ϕ measures the difference between the function value and its local linear part. Thus a change in rational shape changes both the features and the way matrix perturbations propagate through them.

For an input perturbation v and an output sensitivity b , the two Jacobian actions are

$$Jv = d_\phi \odot v + r_\phi \frac{u^\top v}{w}, \quad J^\top b = d_\phi \odot b + u \frac{r_\phi^\top b}{w}. \quad (9)$$

These expressions describe the incoming path through the activation and its backward sensitivity. The outgoing path acts directly on $h = \Phi(z)$. This distinction determines the two response summaries used in the optimizer. The derivatives hold at differentiability points; at zero arguments the absolute-value terms use the selected autodifferentiation derivative $\text{sign}(0) = 0$.

The positive denominator ensures that finite learned coefficients introduce no real poles. All groups begin from a common SiLU approximation and then learn their own nonlinear responses; the initialization is specified with the training algorithm in Appendix D.

B RESPONSE-GUIDED MATRIX DIRECTIONS

TILLER constructs an update orientation from two views of the same momentum. The Muon polar transformation rebalances its singular values; factorized adaptation rescales entries using their gradient history. The learned activation determines how far to rotate from the first direction toward the second. Keeping the norm fixed separates this orientation choice from the subsequent allocation of strength across groups.

Let $g_{t,i}^r$ be the accumulated, distributed-averaged, globally clipped gradient for group $i = (\ell, g)$ and role $r \in \{A, B\}$. With $v_{0,i}^r = 0$, the first-moment buffer and Nesterov momentum are

$$v_{t,i}^r = \beta_1 v_{t-1,i}^r + (1 - \beta_1) g_{t,i}^r, \quad m_{t,i}^r = \beta_1 v_{t,i}^r + (1 - \beta_1) g_{t,i}^r. \quad (10)$$

We apply Muon’s approximate polar map $\mathcal{P}_5$ separately to each group block (Jordan et al., 2024; Liu et al., 2025). For expanded width $H_r = Gw$, the parent direction is

$$p_{t,i}^r = \mu_{\text{mlp}} \chi_{\text{grp}} \mathcal{P}_5(m_{t,i}^r), \quad \mu_{\text{mlp}} = 0.2 \sqrt{\max\{H_r, d\}}, \quad \chi_{\text{grp}} = \sqrt{\frac{\min\{H_r, d\}}{G \min\{w, d\}}}. \quad (11)$$

The factor χ_{grp} places the collection of groupwise polar directions on the nominal Frobenius scale of the full projection. The same calibration applies to the incoming and outgoing roles.

The adaptive direction uses row and column second moments (Shazeer & Stern, 2018). Suppressing the group and role indices, initialize $R_0 = C_0 = 0$ and define

$$\begin{aligned} R_t &= \beta_2 R_{t-1} + (1 - \beta_2) \text{rowsum}(g_t^2), \\ C_t &= \beta_2 C_{t-1} + (1 - \beta_2) \text{colsum}(g_t^2), \\ V_{t,jk} &= \frac{R_{t,j} C_{t,k}}{\max\{\sum_s R_{t,s}, \delta_{\text{fp}}\}(1 - \beta_2^t)}, \quad a_t = \mu_{\text{mlp}} \frac{m_t}{\sqrt{V_t} + \epsilon_{\text{opt}}}. \end{aligned} \quad (12)$$

Here row and column sums are taken over squared entries. We use $\beta_1 = \beta_2 = 0.95$, $\epsilon_{\text{opt}} = 10^{-8}$, and $\delta_{\text{fp}} = \text{finfo}(\text{float32})$. tiny. The use of the same m_t makes the two candidates differ in their matrix transformation rather than in the gradient history they average.

The activation supplies two complementary measurements for choosing between their orientations. Participation measures the breadth of the response, while congruence measures its alignment with the initial response at the current inputs. For a sampled position k , let J_{ki} and h_{ki} denote the group Jacobian and features. The participation ratios are

$$\text{pr}_A(J) = \frac{[\text{tr}(JJ^\top)]^2}{w \|JJ^\top\|_F^2}, \quad \text{pr}_B(h) = \frac{\|h\|_2^4}{w \|h\|_4^4}. \quad (13)$$

For the incoming role, the components are squared singular values of the Jacobian; for the outgoing role, they are squared feature values. A nonzero response concentrated in one component has participation $1/w$, while a response spread equally over m components has participation m/w . This common scale makes breadth comparable between the two paths.

The summaries emphasize positions where the loss is sensitive to the corresponding path. With downstream derivative $\xi_{k\ell}$ at the block output, define

$$\alpha_{ki}^A = \|J_{ki}^\top B_i^\top \xi_{k\ell}\|_2^2 / w, \quad \alpha_{ki}^B = \|\xi_{k\ell}\|_2^2 / d, \quad \pi_i^r = \frac{\sum_k \alpha_{ki}^r \text{pr}_r(\mathcal{R}_{ki}^r)}{\sum_k \alpha_{ki}^r}, \quad (14)$$

where $\mathcal{R}_{ki}^A = J_{ki}$ and $\mathcal{R}_{ki}^B = h_{ki}$. For congruence, set $\Gamma_{ki}^A = J_{ki} J_{ki}^\top$ and $\Gamma_{ki}^B = h_{ki} h_{ki}^\top$. A superscript 0 denotes the same quantity evaluated using the initial rational coefficients and the current input and RMS scale. Then

$$\kappa_i^r = \frac{\sum_k \alpha_{ki}^r \langle \Gamma_{ki}^r, \Gamma_{ki}^{r,0} \rangle}{\sqrt{\left(\sum_k \alpha_{ki}^r \|\Gamma_{ki}^r\|_F^2\right) \left(\sum_k \alpha_{ki}^r \|\Gamma_{ki}^{r,0}\|_F^2\right)}}. \quad (15)$$

Congruence is a cosine between weighted collections of response kernels. Evaluating both collections at the same input isolates the change in activation shape from a change in the input distribution. The kernels compare how responses distribute energy; a common positive rescaling leaves their congruence equal to one.

These definitions give the rotation fraction $e_i^r = \pi_i^r (1 - (\kappa_i^r)^2)$. An unchanged response selects the Muon orientation. Departure from that response permits a rotation, and participation moderates it when the response is concentrated in few components. The bounds follow from the geometry of the summaries: for nonnegative v_j , $\sum_j v_j^2 \leq (\sum_j v_j)^2 \leq w \sum_j v_j^2$, so nonzero participation lies in $[1/w, 1]$. The congruence numerator is nonnegative, since $\langle JJ^\top, J^0 J^{0\top} \rangle = \|J^\top J^0\|_F^2$ and $\langle hh^\top, h^0 h^{0\top} \rangle = (h^\top h^0)^2$. Cauchy–Schwarz bounds the cosine above by one. Consequently $e_i^r \in [0, 1]$.

Suppressing the role index again, remove the parallel component of a_i and orient the remainder using the current gradient:

$$a_i^\perp = a_i - \frac{\langle a_i, p_i \rangle}{\|p_i\|_F^2} p_i, \quad \tilde{a}_i = \sqrt{1 - e_i} p_i + \sigma_i \sqrt{e_i} \frac{\|p_i\|_F}{\|a_i^\perp\|_F} a_i^\perp, \quad (16)$$

where $\sigma_i = 1$ if $\langle g_i, a_i^\perp \rangle \geq 0$ and -1 otherwise. Writing $e_i = \sin^2 \vartheta_i$ identifies this operation as a rotation through ϑ_i in the plane of the two orthogonal directions. In particular,

$$\|\tilde{q}_i\|_F^2 = (1 - e_i) \|p_i\|_F^2 + e_i \|p_i\|_F^2 = \|p_i\|_F^2.$$

The tangent contribution has nonnegative contraction with g_i . We select $q_i = \tilde{q}_i$ when it is finite and both $\langle g_i, p_i \rangle > 0$ and $\langle g_i, \tilde{q}_i \rangle > 0$; otherwise $q_i = p_i$. A zero parent or tangent also selects p_i . Thus the orientation choice preserves the parent norm in either case.

Attention uses this construction with a layer-level response summary. Let $\bar{\pi}_\ell^r = G^{-1} \sum_g \pi_{\ell g}^r$, and obtain κ_ℓ^r by summing the numerator and the two norm sums in equation 15 across groups before normalization. The rotation fraction for the QKV and attention-output matrices is

$$e_\ell^{\text{att}} = \sqrt{\bar{\pi}_\ell^A \bar{\pi}_\ell^B} (1 - \kappa_\ell^A \kappa_\ell^B). \quad (17)$$

It combines the breadth of both feed-forward response paths, and a change in either path can affect the rotation. The polar map groups the 16 attention heads into blocks of four; the stacked QKV direction is scaled by $1/\sqrt{3}$ for its three projections. Factorized moments for the adaptive tangent are computed at the matrix level. The global coefficients derived next apply to the paired GRAIN projections; attention receives its response-guided rotation directly.

C JOINT LOSS-RESPONSE COORDINATION

The direction construction determines the orientation of each group update. Coordination determines how these directions act together. Disjoint parameter blocks can make reinforcing or offsetting contributions to the loss. TILLER records those contributions at shared activation positions and uses their joint second moment to redistribute a fixed update norm.

C.1 SHARED RESPONSE MEASUREMENTS AND THE LOSS LEDGER

For probe k and group $i = (\ell, g)$, perturb both projections along (q_i^A, q_i^B) . Differentiating $(B_i + \alpha q_i^B) \Phi_i((A_i + \alpha q_i^A) x_{k\ell})$ at $\alpha = 0$ gives

$$\delta y_{ki} = B_i J_{ki} q_i^A x_{k\ell} + q_i^B h_{ki}, \quad S_{t,ki} = \langle q_i^A x_{k\ell}, J_{ki}^\top B_i^\top \xi_{k\ell} \rangle + \langle q_i^B h_{ki}, \xi_{k\ell} \rangle. \quad (18)$$

Probe sensitivities and parameter gradients must describe the same loss. If a backward pass averages over R positions per microbatch and M accumulated microbatches, its captured derivative $\bar{\xi}_{k\ell}$ is converted to

$$\xi_{k\ell} = RM \chi_{\text{clip}} \bar{\xi}_{k\ell}, \quad (19)$$

where χ_{clip} is the realized global clipping factor. This restores the summed-loss scale and applies the same clipping as the parameter gradients. The clipping factor is held fixed in the directional measurements.

The two terms follow the incoming and outgoing paths through the block. Each score is a position's contribution to the directional loss derivative. Averaging all such contributions on the scale in Equation 19 recovers the clipped minibatch derivative. The exact current derivative for each paired direction is available directly from the parameter gradients:

$$s_{t,i} = \langle g_{t,i}^A, q_{t,i}^A \rangle + \langle g_{t,i}^B, q_{t,i}^B \rangle. \quad (20)$$

TILLER uses s_t for current descent and the probe matrix $S_t \in \mathbb{R}^{K \times C}$ for the variation of contributions across shared positions. These quantities answer different parts of the coordination problem.

For coefficients c , the vector $S_t c$ contains the combined probe responses. The cross products in $\|S_t c\|_2^2$ distinguish directions with identical mean descent but different overlap. For example, the columns $(2, 0)^\top$ and $(0, 2)^\top$ each have mean one and their sum has mean squared response four. Two identical columns $(2, 0)^\top$ retain the same individual means but produce a mean squared response of eight when added. The ledger captures this difference, allowing the coefficient model to favor combinations with less concentrated joint response for a given descent signal.

Weight decay contributes another direction to the same perturbation. Evaluating equation 18 with $(\lambda_{\text{wd}}A_i, \lambda_{\text{wd}}B_i)$ and summing over the GRAIN groups gives a vector $a_t^{\text{dec}} \in \mathbb{R}^K$. The response penalty then expands as

$$\frac{1}{K} \|S_t c + a_t^{\text{dec}}\|_2^2 = c^\top \frac{S_t^\top S_t}{K} c + 2c^\top \frac{S_t^\top a_t^{\text{dec}}}{K} + \frac{\|a_t^{\text{dec}}\|_2^2}{K}. \quad (21)$$

The final term is independent of c . The first two terms supply the response metric and its cross term with decay.

TILLER refreshes $K = 32$ shared probes at steps 1, 9, 17, $\dots$. The positions coincide across layers, and each refresh recomputes the participation, congruence, and response scores at current parameters. Between refreshes, it reuses the response summaries and updates the ledger along the current gradient contraction. Let $\nu = \|S_{\text{ref}}\|_F / \sqrt{K}$ be the RMS row norm of the latest refresh. The intervening rows are

$$S_t = \mathbf{1}_K (\gamma_t s_t)^\top, \quad a_t^{\text{dec}} = \mathbf{1}_K \gamma_t a_t^{\text{dec}}, \quad \gamma_t = \begin{cases} \nu / \|s_t\|_2, & \|s_t\|_2 > 0, \\ 0, & \|s_t\|_2 = 0, \end{cases} \quad (22)$$

where a_t^{dec} is the current gradient contraction with the GRAIN decay direction. For $s_t \neq 0$, these rows have the same second-moment trace as the most recent probes: $\|S_t\|_F^2 / K = \nu^2$. They supply current orientation information along s_t while probe refreshes recover variation across positions.

The uncompressed ledger would follow

$$G_t^* = \beta_2 G_{t-1}^* + (1 - \beta_2) S_t^\top S_t / K, \quad G_1^* = S_1^\top S_1 / K. \quad (23)$$

For any c , $c^\top G_t^* c$ is a discounted average of squared responses for that coefficient combination. The measurements retain the directions used when they were recorded; discounting emphasizes the response geometry encountered more recently.

We represent this history by a factor $\mathcal{B}_t$ and an isotropic remainder $\tau_t I$. New rows enter the current metric before compression:

$$\mathcal{A}_t = \begin{bmatrix} \sqrt{\beta_2} \mathcal{B}_{t-1} \\ \sqrt{1 - \beta_2} S_t \end{bmatrix}, \quad G_t = \mathcal{A}_t^\top \mathcal{A}_t / K + \beta_2 \tau_{t-1} I, \quad r_t = \beta_2 r_{t-1} + (1 - \beta_2) S_t^\top a_t^{\text{dec}} / K. \quad (24)$$

At the first step, $\mathcal{A}_1 = \mathcal{B}_1 = S_1$, $\tau_0 = \tau_1 = 0$, and $r_1 = S_1^\top a_1^{\text{dec}} / K$. The current coefficient model therefore uses fresh rows at full resolution together with the compressed history.

Robust Frequent Directions carries this history forward (Ghashami et al., 2016; Luo et al., 2019). If the singular values of $\mathcal{A}_t$ are $\sigma_1 \geq \sigma_2 \geq \dots$, set $\Delta_t = \sigma_{r+1}^2$ for $r = 64$ and subtract Δ_t from each retained squared singular value. When the rank is at most r , take $\Delta_t = 0$ and retain an exact factor. With right singular vectors $V_{1:r}$, the compressed representation is

$$\mathcal{B}_t = \text{Diag}(\sqrt{\sigma_1^2 - \Delta_t}, \dots, \sqrt{\sigma_r^2 - \Delta_t}) V_{1:r}^\top, \quad \tau_t = \beta_2 \tau_{t-1} + \frac{\Delta_t}{2K}. \quad (25)$$

The factor retains dominant combinations of responses. The isotropic term assigns energy to directions outside its span. The midpoint $\Delta_t / (2K)$ balances the error introduced by spectral shrinkage, as the following bound shows.

Proposition 1 (Response-metric approximation). *For $K > 0$, $\beta_2 \in [0, 1]$, and $\tau_{t-1} \geq 0$, let G_t and $(\mathcal{B}_t, \tau_t)$ be defined by equations 24 and 25. Then $\hat{G}_t = \mathcal{B}_t^\top \mathcal{B}_t / K + \tau_t I$ is positive semidefinite and*

$$\|\hat{G}_t - G_t\|_2 \leq \frac{\Delta_t}{2K}, \quad |c^\top (\hat{G}_t - G_t) c| \leq \frac{\Delta_t}{2K} \|c\|_2^2 \quad \text{for every } c \in \mathbb{R}^C. \quad (26)$$

Proof. Both terms of $\hat{G}_t$ are positive semidefinite. When $\Delta_t = 0$, the metrics coincide. Otherwise use an orthonormal basis of right singular vectors of $\mathcal{A}_t$, completed by its null space. On a retained direction, shrinkage removes Δ_t / K and the isotropic term restores $\Delta_t / (2K)$, giving error $-\Delta_t / (2K)$. On a discarded direction with squared singular value $s \in [0, \Delta_t]$, the error is $(\Delta_t / 2 - s) / K$; on a null direction it is $\Delta_t / (2K)$. Each absolute eigenvalue error is at most $\Delta_t / (2K)$. The quadratic-form bound follows from $|c^\top E c| \leq \|E\|_2 \|c\|_2^2$ for symmetric E . $\square$

The bound also describes accumulation over the discounted history. If $E_t = \hat{G}_t - G_t^*$, each compression adds an error D_t with $\|D_t\|_2 \leq \Delta_t/(2K)$, so $E_t = \beta_2 E_{t-1} + D_t$ and $E_1 = 0$. Applying the triangle inequality recursively gives

$$\|E_t\|_2 \leq \sum_{j=2}^t \beta_2^{t-j} \frac{\Delta_j}{2K}. \quad (27)$$

Older compression errors are attenuated on the same time scale as older response measurements.

C.2 EQUAL-BUDGET COEFFICIENT SELECTION

The ledger becomes useful for optimization when combined with the current descent signal. For group energy $\omega_i = \|q_i^A\|_F^2 + \|q_i^B\|_F^2$ and $\mathcal{E}_0 = \sum_i \omega_i$, the disjoint parameter blocks give $\|\Delta\theta(c)\|_2^2 = \eta^2 \sum_i \omega_i c_i^2$. Constraining this sum to $\eta^2 \mathcal{E}_0$ preserves the norm of the uncoordinated structured update. The model in equation 5 then trades current descent against the ledger’s joint response penalty. Its coefficients can be signed because the model assesses the combination as a whole.

For positive ω_i , write $W = \text{Diag}(\omega)$ and $y = W^{1/2}c$. Dividing the objective by η^2 gives

$$\begin{aligned} \mathcal{Q}_t(y) &= \frac{1}{2} y^\top M_t y - b_t^\top y, \quad \|y\|_2^2 = \mathcal{E}_0, \\ M_t &= W^{-1/2} G_t W^{-1/2}, \quad b_t = W^{-1/2} (s_t/\eta - r_t). \end{aligned} \quad (28)$$

This change of coordinates turns the weighted norm constraint into a sphere. We use the positive-multiplier branch of its stationarity equation to construct a proposal. The stationarity equation is $(M_t + \lambda I)y = b_t$, where the multiplier λ controls the norm. TILLER constructs a proposal in a subspace generated by b_t and repeated metric actions. A reorthogonalized Lanczos recurrence (Lanczos, 1950) produces an orthonormal basis Q_k , with $k \leq 32$. Each action uses the ledger factor:

$$M_t v = W^{-1/2} \mathcal{A}_t^\top (\mathcal{A}_t W^{-1/2} v) / K + \beta_2 \tau_{t-1} W^{-1} v. \quad (29)$$

Let T_k^+ be the positive-semidefinite projection of the accumulated Lanczos matrix. For $y = Q_k z$, diagonalize $T_k^+ = U \text{Diag}(\theta) U^\top$ and set $\gamma = U^\top Q_k^\top b_t$. The projected stationarity and norm conditions reduce to

$$z = U \text{Diag}((\theta + \lambda)^{-1}) \gamma, \quad \psi(\lambda) = \sum_j \frac{\gamma_j^2}{(\theta_j + \lambda)^2} = \mathcal{E}_0. \quad (30)$$

For nonzero γ and $\lambda > 0$, $\psi'(\lambda) = -2 \sum_j \gamma_j^2 / (\theta_j + \lambda)^3 < 0$. The positive branch therefore has at most one root. We bracket it by

$$\lambda_{\text{lo}} = \epsilon_{\text{mach}} C, \quad \lambda_{\text{hi}} = \|b_t\|_2 / \sqrt{\mathcal{E}_0} + \lambda_{\max}(T_k^+) + \lambda_{\text{lo}}. \quad (31)$$

Since $\theta_j \geq 0$ and Q_k is orthonormal, $\psi(\lambda_{\text{hi}}) \leq \|b_t\|_2^2 / \lambda_{\text{hi}}^2 \leq \mathcal{E}_0$. If $\psi(\lambda_{\text{lo}}) \geq \mathcal{E}_0$, 64 bisection iterations give the proposal $c = W^{-1/2} Q_k z$. Otherwise the step retains $c = \mathbf{1}$.

The projected solution is a candidate for the full ledger model. TILLER evaluates it in F_t and selects it only if it improves on $\mathbf{1}$, is finite, and satisfies

$$\frac{|c^\top W c - \mathcal{E}_0|}{\max\{\mathcal{E}_0, 1\}} \leq 10^{-8}. \quad (32)$$

For each layer and each matrix role it also checks

$$\sum_g c_{\ell g} \langle g_{\ell g}^r, q_{\ell g}^r \rangle > 0, \quad \sum_g c_{\ell g} \langle m_{\ell g}^r, q_{\ell g}^r \rangle > 0, \quad r \in \{A, B\}. \quad (33)$$

These contractions are checked for both the candidate and $\mathbf{1}$. A failed check, a zero coordinate energy, or a degenerate coefficient problem selects $\mathbf{1}$. The result is an allocation evaluated against the complete response model, with the projected solve used to construct the proposal.

D FULL AND HYBRID TRAINING ALGORITHMS

Table 1 assembles the activation, direction, and coordination operations into a training algorithm. A switch time $T_s = T$ gives full TILLER. A switch time $T_s < T$ gives TILLER–Muon: the model follows the same TILLER updates through step T_s and then continues with ordinary matrix-level Muon. The shared prefix and explicit state transfer define the relation between the two variants.

The transition changes the matrix-direction construction. During the TILLER stage, feed-forward directions use groupwise polar maps, response-guided rotations, and joint coefficients. During the Muon stage, each full incoming, outgoing, QKV, and attention-output matrix receives an ordinary Muon update. Copying the first-moment buffers preserves gradient history across this change. The AdamW optimizer for the remaining parameters is the same object throughout, so its moments and step counts continue. The switch is applied after TILLER update T_s ; update $T_s + 1$ is the first Muon update. The learning-rate schedule advances with the global training step. Rational coefficients continue learning through AdamW in both stages.

The denominator in equation 2 is at least one, so finite learned coefficients introduce no real poles. Each group starts from the same SiLU fit on $[-5, 5]$. The numerator and denominator coefficients, in ascending powers, are

$$(a_0, \dots, a_5) = (2.4915714620520875 \times 10^{-5}, 0.5000668663485823, 0.2499441908995771, 0.0526200910151016, 0.005525765640115002, 0.00024199321178747251), \quad (34)$$

$$(b_1, \dots, b_4) = (0.00022554017910995938, 0.10511420350691994, 2.8656992394109812 \times 10^{-5}, 0.0004816962110562025). \quad (35)$$

The ten coefficients then learn independently in every group. We use group width $w = 256$ and $\epsilon_{\text{act}} = 10^{-6}$.

Table 1: GRAIN training with full TILLER or TILLER–Muon. The full schedule uses $T_s = T$; the hybrid uses $T_s < T$. Both retain the learned activation throughout training.

Training procedure	
1	Given a training horizon T , switch point $T_s \in \{1, \dots, T\}$, and learning-rate schedule $\{\eta_t\}_{t=1}^T$, initialize the Transformer with GRAIN; initialize momentum, factorized moments, AdamW state, and an empty loss ledger.
2	For $t = 1, \dots, T$:
3	Evaluate the minibatch loss, backpropagate, and clip the accumulated gradients. Capture shared probes during the pass when $t \leq T_s$ and $t \in \{1, 9, 17, \dots\}$.
4	If $t \leq T_s$:
5	Update the first and factorized second moments; form the parent Muon and adaptive directions (equations 10–12).
6	On a probe refresh, recompute participation and congruence. Use the latest summaries to construct the response-guided GRAIN and attention directions (equations 16 and 17).
7	Compute s_t . Form the probe scores and decay responses on a refresh; otherwise use the trace-matched rows in equation 22.
8	Form G_t, r_t from the ledger and new rows. Propose and select c using Appendix C.2; compress the ledger for the next step using equation 25.
9	Apply $A_i \leftarrow (1 - \eta_t \lambda_{\text{wd}})A_i - \eta_t c_i q_i^A$ and $B_i \leftarrow (1 - \eta_t \lambda_{\text{wd}})B_i - \eta_t c_i q_i^B$. Apply the response-guided attention updates.
10	Else:
11	Apply ordinary Muon to each full feed-forward and attention matrix using its retained momentum buffer.
12	End if.
13	Update rational coefficients, embeddings, normalization parameters, and remaining tensors with AdamW.
14	If $t = T_s < T$, transfer each structural matrix’s first-moment buffer to Muon and remove TILLER response hooks. Retain model parameters, AdamW state, and the global schedule.
15	End for.

The polar map uses five Newton–Schulz iterations. It transposes tall inputs, casts the working matrix to bfloat16, and normalizes by a Frobenius norm accumulated in float32, lower bounded by 10^{-7} , and cast to bfloat16. With $(a, b, c) = (3.4445, -4.775, 2.0315)$, the iteration is

$$X_{j+1} = aX_j + b(X_jX_j^\top)X_j + c(X_jX_j^\top)^2X_j. \quad (36)$$

The original orientation is restored after the final iteration. Factorized second moments are maintained in float32. The coefficient solver uses double precision and two reorthogonalization passes per Lanczos basis vector. At an early breakdown, it reorthogonalizes a deterministic cosine vector to start a new block with zero coupling to the preceding block. The resulting basis is checked for orthogonality, and negative eigenvalues of the accumulated projected matrix are set to zero before the multiplier solve.

For a zero response, participation is defined as zero. If either matrix role has zero total sensitivity, both participation values are set to one. If either congruence denominator vanishes, both congruence values are set to one. Congruence is also one when the rational coefficients equal their initialization. These conventions give zero departure when the response comparison is undefined.